A2.2 Cell structure

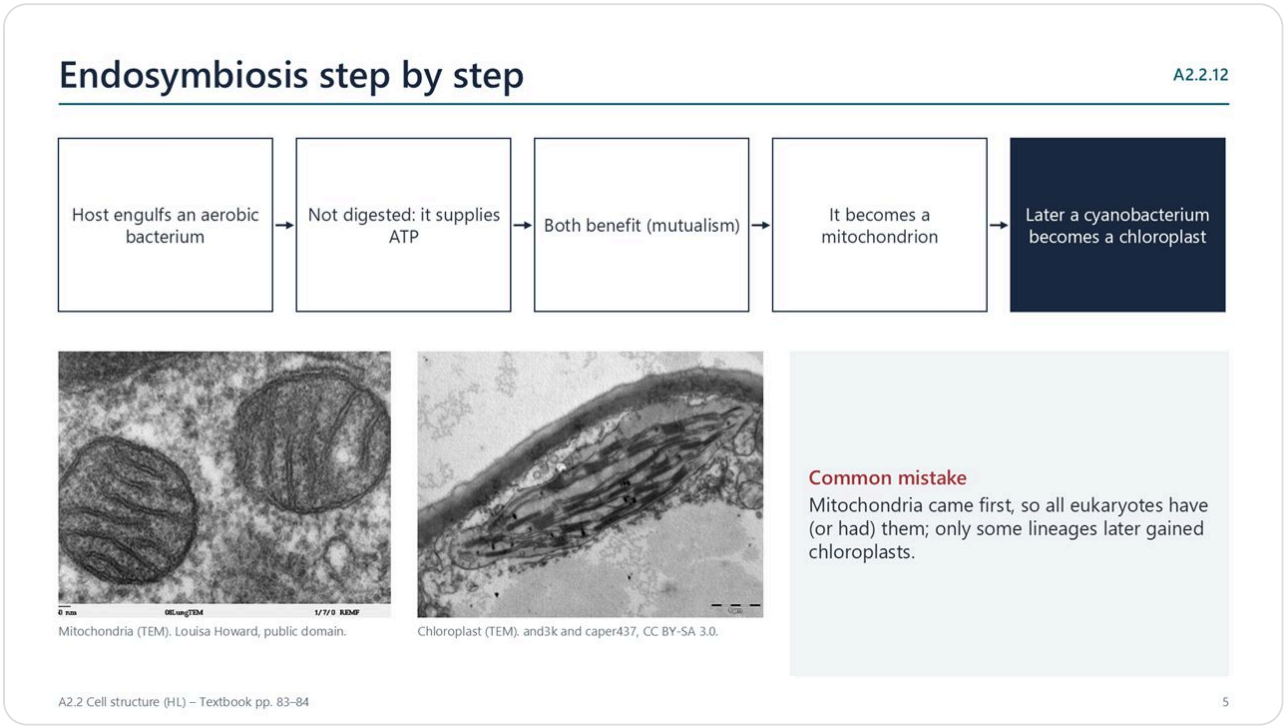
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-a2-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 82–87 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Where eukaryotic cells came from

A2.2.12 Endosymbiosis

pp. 82–85



Mitochondria (TEM). Louisa Howard, public domain.

Chloroplast (TEM). and3k and caper437, CC BY-SA 3.0.

All eukaryotes probably share one unicellular ancestor that already had a nucleus and reproduced sexually. A host cell took in an aerobic bacterium by endocytosis but did not digest it. The two came to depend on each other (mutualism), and over many generations the bacterium became a **mitochondrion**. Later, in the ancestor of algae and plants, an engulfed cyanobacterium became a **chloroplast**. **Evidence:** both organelles have **70S ribosomes**, **naked circular DNA** and a double membrane, are about the size of bacteria, and arise only by binary fission of existing organelles. Their DNA sequences are closest to alphaproteobacteria (mitochondria) and cyanobacteria (chloroplasts). The theory made predictions that sequencing later confirmed.

Many cells, many roles

A2.2.13 Differentiation

pp. 85–86

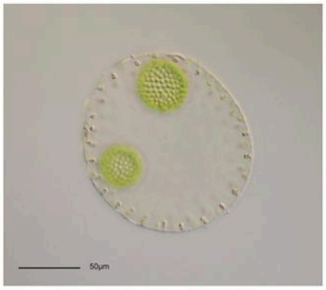
All the cells of an organism come from one zygote by mitosis, so they share the same **genome**. During **differentiation** different genes are expressed in different cells, triggered by signals from the environment and by the cell's position. Each cell type therefore has its own **proteome**: a developing red blood cell makes large amounts of haemoglobin, a muscle cell makes actin and myosin. Specialized cells form tissues and organs and share the work, but each depends on the others.

A2.2.14 The evolution of multicellularity

p. 87

Multicellularity evolved many times

A2.2.14



Volvox colonies with daughter colonies inside.
Picturepest, CC BY 2.0.

Animals, plants, many fungi and many algae are multicellular, but these groups are far apart on the tree of life: multicellularity arose **independently** many times.

Volvox: a ball of flagellated cells with a few reproductive cells.

Advantages	Requirements
Larger size; new niches	Cells stick together
Specialized cells work efficiently	Cells signal and differentiate

A2.2 Cell structure (HL) – Textbook p. 87 16

All animals and plants, many fungi and many algae are multicellular. These groups are separated on the tree of life by unicellular ones, so multicellularity evolved **independently many times**. It allows a larger body and **specialized cells**, but needs cells that stick together, signal to each other and differentiate. Volvox colonies and choanoflagellates, the closest living relatives of animals, show how the change may have begun.

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